

07/10/20

Day - 9

kubernetes file  $\Rightarrow$  resource object.

documentation  $\Rightarrow$  kubernats.

minikube version

minikube start  $\Rightarrow$  download packages.

minikube kubectl -- get nodes  $\Rightarrow$  It will be show what clusters byt.

get command

minikube kubectl -- get ~~nodes~~ pods -A

default - a each container - the pod create and run again

minikube kubectl -- run apache pod --image = httpd

is cmd used for create pod.

Pod name: ... image name

minikube kubectl -- get pods  $\Rightarrow$  get pods = 0, no crash

minikube kubectl -- describe pod apache  $\Rightarrow$  Suppose pod created it will be default.

It will be show complete metadata about pod

minikube kubectl -- apply -f simplepod.yaml

minikube kubectl -- get pods

minikube kubectl -- delete -f simplepod.yaml

minikube kubectl -- get pods

Two ways to create subnetworks:

1. Declarative method  
on this method the resources mentioned in resource object file will be created.
2. Imperative method

kubectl commands are used to create a resource.

K8s => what purpose to write yaml file.

you should mention this

Declarative method: to create pod.

notepad simplepod.yaml

copy the file to paste this file

minikube kubectl -- apply -f simplepod.yaml => created

minikube kubectl -- get pods

minikube kubectl -- delete -f simplepod.yaml => pod delete

Deployment:

TO implement replica set => use deployment.

Deployment strategy => Rollback => Implement parallel undo parallel.  
Rollout => deployment a. Implement parallel  
Rolling up => use deployment resources

minikube kubectl -- create deployment demo deploy --image=python

minikube kubectl -- get deploy

minikube kubectl -- get pods

Deployment  
name

--image=python:2.7

python img not  
working

How to horizontally pods

minikube kubectl -- scale deployment nginxdeploy --replicas=5  
scale and use parri scale up and scale down. parri must use.

minikube kubectl -- get deploy -A

create  
5 pods  
no. of pods

check <sup>created</sup> pod image version

minikube kubectl -- describe deploy demo deploy.

change the image version. - and you can also change the image  
update the image.

minikube kubectl -- set image deployment/nginxdeploy nginx=nginx:1.30.5

change the image

minikube kubectl -- set image deployment/nginxdeploy nginx=httpd

delete pods manually

minikube kubectl -- delete pod pod names

amma pods delete parra udane amma next deploy  
check pannum patha again pods <sup>automatic</sup> create again. => because amma  
deploy kulla iruvathurala athoda work crash ana pods-ku  
patha again vera pods create pannum.  
deploy delete pannita => pods create again

rolling out - status check.

minikube kubectl -- rollout status deployment nginxdeploy

rollback => go to previous state

minikube kubectl -- rollout undo deployment nginxdeploy

Task  $\Rightarrow$  deployment  $\rightarrow$  resource file create  $\rightarrow$  then apply using `kubectl`

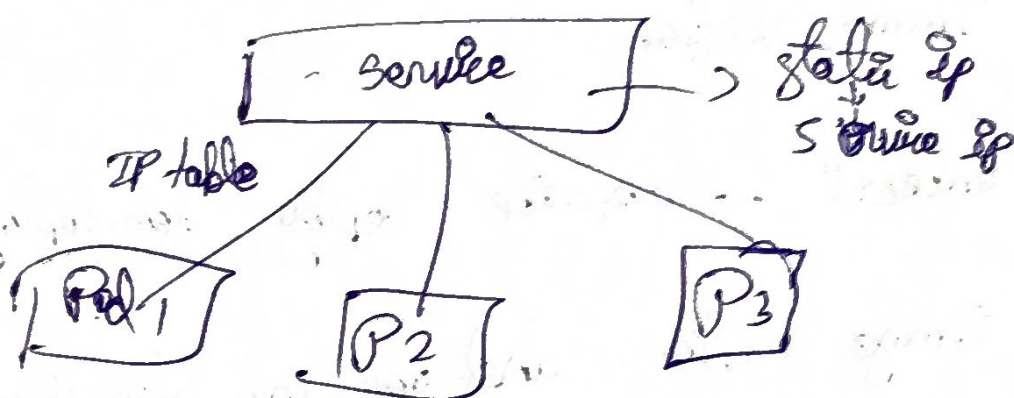
## Network:

static IP  $\Rightarrow$  use service  
need



static IP that IP service

IP table  $\Rightarrow$  proxy  $\rightarrow$  passing



CNI = container network interface

CNI  $\Rightarrow$  has a particular IP



It helps same node-to network pod to pod communication

node to node communication  $\Rightarrow$  use route table

Kubernetes cluster form  $\Rightarrow$  use n/w policy

Some IP-to communicate same path same conflict  $\Rightarrow$  to avoid this use different port, same IP

### service

1. cluster IP  $\Rightarrow$  default is already used to be  $\Rightarrow$  Kubernetes  $\rightarrow$  create
2. Node port  $\Rightarrow$  it helps for port expose
3. Load Balancer
4. External name